

Diogo M. Camacho

+1 617 945 4383 | diogo.camacho.2008@gmail.com | [diogocamacho](#) | [diogocamacho](#)

Executive Profile

Led and built computational organizations across biotech and pharma for 15+ years, growing early-stage ventures into scaled platform teams. Expert in ML, multi-omics, and computational biology, with working command of the full discovery stack through preclinical validation. Drove organization-wide AI/ML adoption, built the computational platform narrative for fundraising and pharma partnerships (Roche, Novo Nordisk), and presented strategy to boards and VC partners. Led global teams across the US and Europe. Secured >\$2.4M in competitive DARPA and NIH funding; 6 patents; 20+ publications (7,000+ citations).

- **Leadership through ambiguity:** Built and scaled computational organizations across evolving ventures, from an undefined early mandate to a functioning platform team. Comfortable owning technical direction before the path is clear.
- **Technical innovation:** Established AI platforms across the discovery lifecycle: patient-stratification, active-learning frameworks, and agentic systems spanning target ID through preclinical validation.
- **Building and judging talent:** Recruit and develop computational leaders: hiring, mentorship, succession planning, and the executive searches that fill senior gaps. Built and held a 25-person group at Cellarity with 90%+ retention.
- **Executive credibility:** Board and VC interface across multiple companies. Secured >\$2.4M in competitive DARPA and NIH funding; 6 patents and 20+ publications (7,000+ citations) across computational biology, ML, and synthetic biology.

Professional Experience

Flagship Pioneering

Cambridge, MA

SENIOR PRINCIPAL, COMPUTATION

Oct 2024 - Present

- **Company origination:** AI lead on the FL111 founding team, owning computational efforts from exploration through seed. Advised FL106, FL108, and FL115 at the same stages.
- **Venture platform strategy:** Set computational and data strategy for multiple seed-stage ventures, building the platform capabilities and technical direction for early programs.
- **Agentic AI systems:** Built agentic AI systems for drug discovery and DTC health applications across the Flagship portfolio.
- **Talent and leadership:** Recruited and retained computational leaders and staff across portfolio companies.

Abiologics (Flagship Pioneering portfolio company)

Cambridge, MA

HEAD OF COMPUTATIONAL SCIENCES

Jan 2025 - Present

- **Platform strategy:** Lead the computational sciences function for a D-amino acid therapeutic platform, owning AI-first design strategy, operations, and execution.
- **Active learning:** Made active learning the engine of mini-binder optimization, cutting time-to-optimized-lead from 10 months to 2 by prioritizing the highest-information candidates each DBT cycle.
- **Agentic AI:** Built and deployed agentic AI systems for target identification, prioritization, and competitive intelligence.
- **Organizational leadership:** Own the function end to end (hiring, performance management, budgeting, executive search) across all 5 therapeutic programs.

Cellarity

Boston, MA

VICE PRESIDENT, COMPUTATIONAL AND DATA SCIENCES

Jul 2022 - Feb 2024

- **Executive leadership:** On the R&D Leadership Team, reporting to the CSO. Shaped corporate strategy connecting the computational platform to therapeutic portfolio direction.
- **Organizational build:** Scaled and led a 25-person computational organization across bioinformatics, ML, and computational biology (Boston and Germany), retaining 95% through a company reorganization.
- **Platform strategy:** Directed platform strategy and operations end to end across all 6 therapeutic programs; the platform delivered the targets behind Cellarity's MASH partnership with Novo Nordisk (up to \$532M).
- **External engagement:** Primary computational interface for the board, founding VC partners, and external collaborators on platform differentiation and roadmap.

Rheos Medicines

Boston, MA

SENIOR DIRECTOR, COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

Nov 2020 - Jul 2022

- **Precision medicine strategy:** Set the computational precision-medicine strategy for immunometabolism drug discovery, defining how patient stratification and multi-omics data drove program decisions.
- **AI and data strategy:** Defined and implemented Rheos' AI and data strategy with Research Informatics across discovery and translational programs.
- **Platform delivery:** Built the MetPM platform and MetKG knowledge graph (patient-stratification and multi-omics tooling adopted across the pipeline), and led the computational team delivering on Rheos's worldwide immunometabolism collaboration with Roche (\$42.5M upfront, up to ~\$750M in milestones).
- **Board engagement:** Presented computational strategy to the board and pharma partners; contributed to the Series B investor narrative.

Wyss Institute @ Harvard University

Boston, MA

LEAD, PREDICTIVE BIOANALYTICS

Jul 2016 - Nov 2020

- **Founding leadership:** Co-founded and led the Predictive BioAnalytics Initiative, a new ML/AI function in translational science: set the research strategy, secured funding, and built the team from scratch.
- **Capital acquisition:** Raised >\$2.4M in competitive DARPA and NIH grants as Co-PI.
- **Infectious disease research:** Co-led the computational side of host immune-tolerance research: transcriptomics that identified infection-response pathways, later validated and published in *Advanced Science*.
- **Team leadership:** Directed and mentored interdisciplinary teams (staff scientists, postdocs, graduate students, interns) across synthetic biology, systems biology, and ML.

Evelo Biosciences

Cambridge, MA

SENIOR SCIENTIST, COMPUTATIONAL SYSTEMS BIOLOGY LEAD

Jan 2015 - Apr 2016

Ember Therapeutics

Cambridge, MA

PRINCIPAL SCIENTIST, COMPUTATIONAL SYSTEMS BIOLOGY LEAD

Jan 2014 - Dec 2014

Pfizer, Inc.

Cambridge, MA

SENIOR RESEARCH SCIENTIST, COMPUTATIONAL SCIENCES CENTER OF EMPHASIS

Jan 2011 - Jan 2014

Howard Hughes Medical Institute @ Boston University

Boston, MA

POSTDOCTORAL FELLOW (JIM COLLINS LAB)

Jul 2007 - Jan 2011

Education

Virginia Polytechnic Institute and State University

Blacksburg, VA

PH.D. IN GENETICS, BIOINFORMATICS, AND COMPUTATIONAL BIOLOGY

Faculdade de Ciencias da Universidade de Lisboa

Lisboa, Portugal

B.Sc. IN BIOCHEMISTRY

Publications and Patents

- Valeri JA, Soenksen LR, Collins KM, Ramesh P, Cai G, Powers R, **Camacho DM**, et al. (2023). BioAutoMATED: An end-to-end automated machine learning tool for explanation and design of biological sequences. *Cell Systems* 14, 525.
- Sperry MM, Novak R, Keshari V, Dinis ALM, Cartwright MJ, **Camacho DM**, Pare JF, Super M, Levin M, Ingber DE (2022). Enhancers of host immune tolerance to bacterial infection discovered using linked computational and experimental approaches. *Advanced Science* 9, 2200222.
- Gazzaniga FS, **Camacho DM**, Wu M, Palazzo MFS, Dinis ALM, Grafton FN, Cartwright MJ, Super M, Kasper DL, Ingber DE (2021). Harnessing colon chip technology to identify commensal bacteria that promote host tolerance to infection. *Frontiers in Cellular and Infection Microbiology* 11, 638014.
- Valeri J, Collins KM, Ramesh P, Alcantar M, Lepe BA, Lu TK, **Camacho DM** (2020). Sequence-to-function deep learning frameworks for engineered riboregulators. *Nature Communications* 11, 5058.
- Bojar D, Powers RK, **Camacho DM**, Collins JJ (2020). Deep-learning resources for studying glycan-mediated host-microbe interactions. *Cell Host & Microbe* 29, 132-144.
- Jalili-Firoozinezhad S, Gazzaniga FS, Calamari EL, **Camacho DM**, Fadel C, Nestor B, Cronce MJ, Tovaglieri A, Levy O, Gregory KE, Breault DT, Cabral JMS, Kasper DL, Novak R, Ingber DE (2019). A complex human gut

- microbiome cultured in an anaerobic intestine-on-a-chip. *Nature Biomedical Engineering* 3, 520-531.
- Tovaglieri A, Sontheimer-Phelps A, Geirnaert A, Prantil-Baun R, **Camacho DM**, Chou DB, Jalili-Firoozinezhad S, de Wouters T, Kasendra M, Super M, Cartwright M, Richmond CA, Breault DT, Lacroix C, Ingber DE (2019). Species-specific enhancement of enterohemorrhagic *E. coli* pathogenesis mediated by microbiome metabolites. *Microbiome* 7, 43.
 - Bojar D, **Camacho DM**, Collins JJ (2019). Using natural language processing to learn the language of glycans. *Nature Communications* (in review).
 - **Camacho DM**, Collins KM, Powers RK, Costello JC, Collins JJ (2018). Next-generation machine learning for biological networks. *Cell* 173, 1581-1592.
 - Musah S, Dimitrakakis N, **Camacho DM**, Church GM, Ingber DE (2018). Directed differentiation of human induced pluripotent stem cells into mature kidney podocytes and establishment of a Glomerulus Chip. *Nature Protocols* 13, 1662-1685.
 - Paandey SP, Winkler JA, Li H, **Camacho DM**, Collins JJ, Walker GC (2014). Central role for RNase YbeY in Hfq-dependent and Hfq-independent small-RNA regulation in bacteria. *BMC Genomics* 15, 121.
 - Galagan JE, Minch K, Peterson M, **Camacho D**, et al. (2013). The Mycobacterium tuberculosis regulatory network and hypoxia. *Nature* 499, 178-183.
 - Belenky P, **Camacho D**, Collins JJ (2013). Fungicidal drugs induce a common oxidative-damage cellular death pathway. *Cell Reports* 3, 350-358.
 - Marbach D, Costello JC, Kuffner R, Vega NM, Prill RJ, **Camacho DM**, Allison KR, Kellis M, Collins JJ, Stolovitzky G (2012). Wisdom of crowds for robust gene network inference. *Nature Methods* 9, 796-804.
 - Dwyer DJ, **Camacho DM**, Callura JM, Kohanski MA, Collins JJ (2011). Antibiotic-induced bacterial cell death exhibits physiological and biochemical hallmarks of apoptosis. *Molecular Cell* 46, 561-572.
 - Modi SR, **Camacho DM**, Kohanski MA, Collins JJ (2011). Functional characterization of bacterial sRNAs using a network biology approach. *Proceedings of the National Academy of Sciences* 108, 15522-15527.
 - **Camacho DM**, Collins JJ (2009). Systems biology strikes gold. *Cell* 137, 24-26.
 - **Camacho D**, Vera-Licona P, Laubenbacher R, Mendes P (2007). Comparison of existing reverse engineering methods by use of an in silico system. *Annals of the New York Academy of Sciences* 1115, 73-89.
 - Mendes P, **Camacho D**, de la Fuente A (2005). Modelling and simulation for metabolomics data analysis. *Biochemical Society Transactions* 33, 1427-1429.
 - **Camacho D**, de la Fuente A, Mendes P (2005). The origin of correlations in metabolomics data. *Metabolomics* 1, 53-63.
 - Martins AM, **Camacho D**, Shuman J, Sha W, Mendes P, Shulaev V (2004). A systems biology study of two distinct growth phases of *Saccharomyces cerevisiae* cultures. *Current Genomics* 5, 649-663.

Patents spanning drug discovery, ML methods, and synthetic biology.

Full publication list available on Google Scholar (7,000+ citations).

Grants

Raised >\$2.4M in competitive DARPA and NIH funding to support translational bioanalytics platforms at the Wyss Institute.